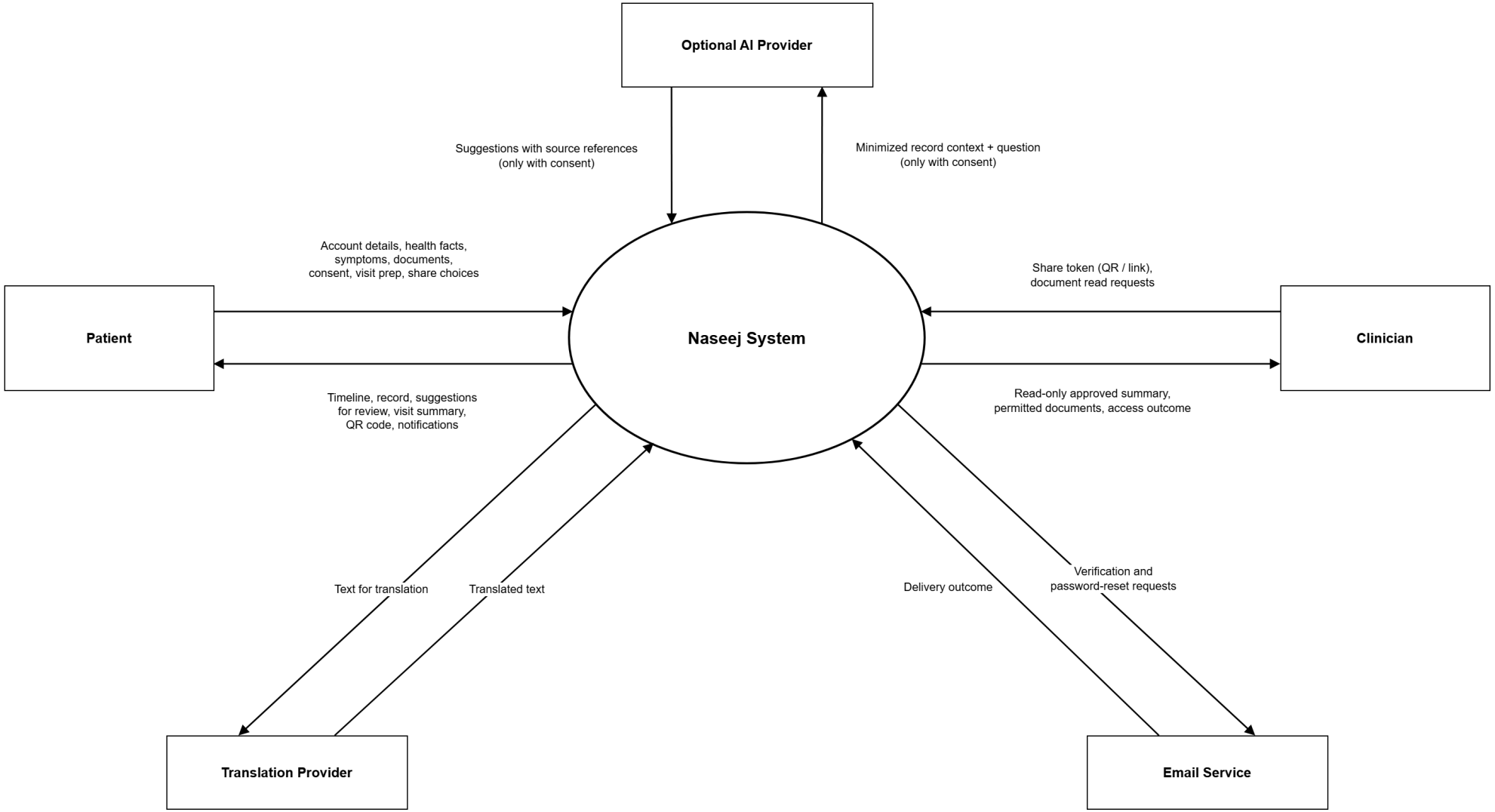
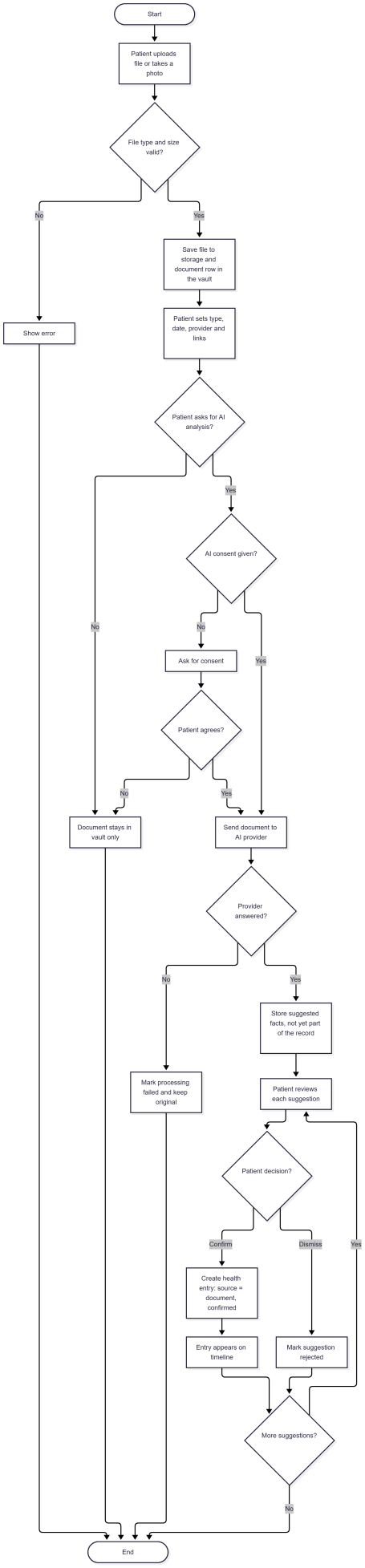
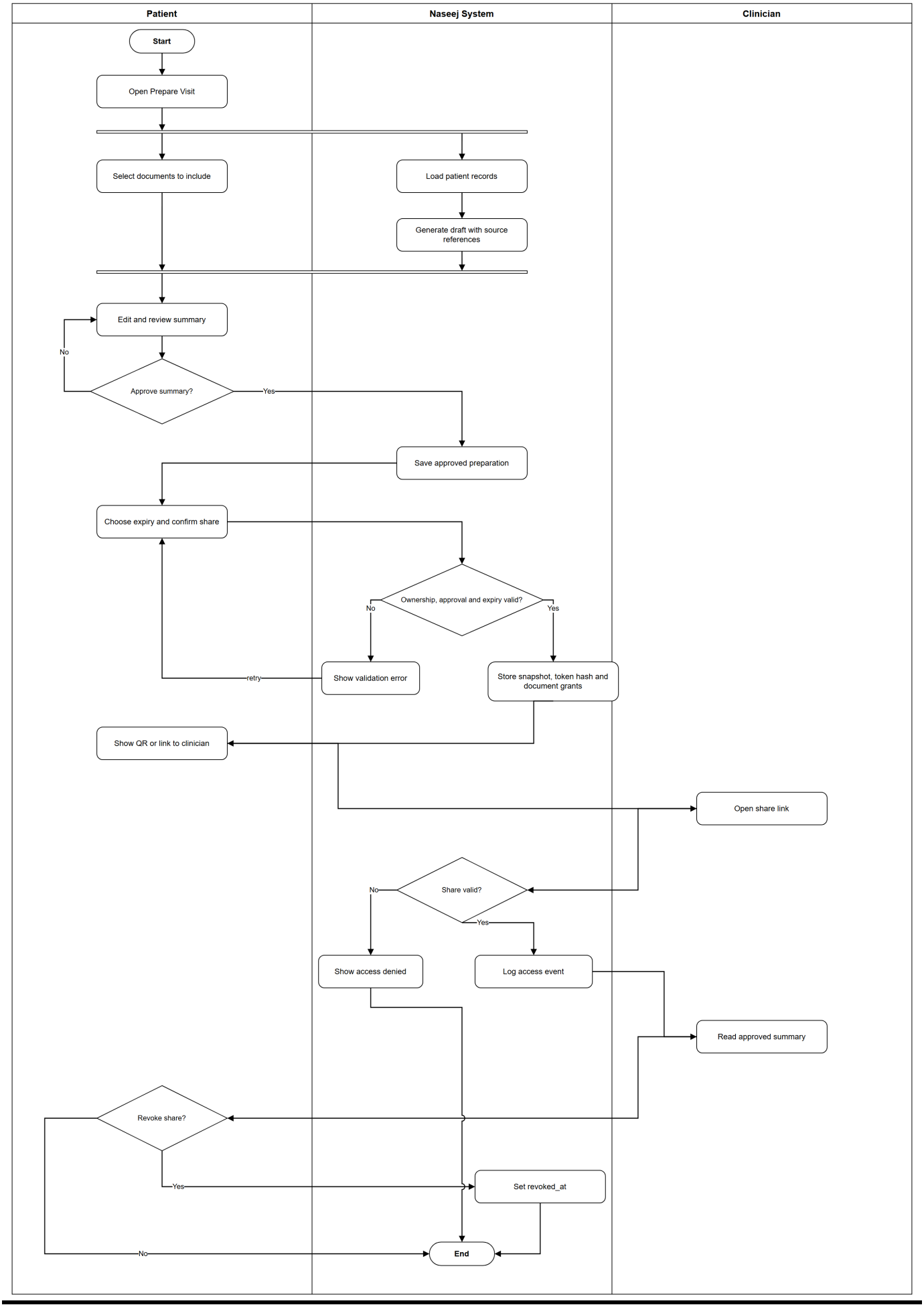
# 1. Context diagram



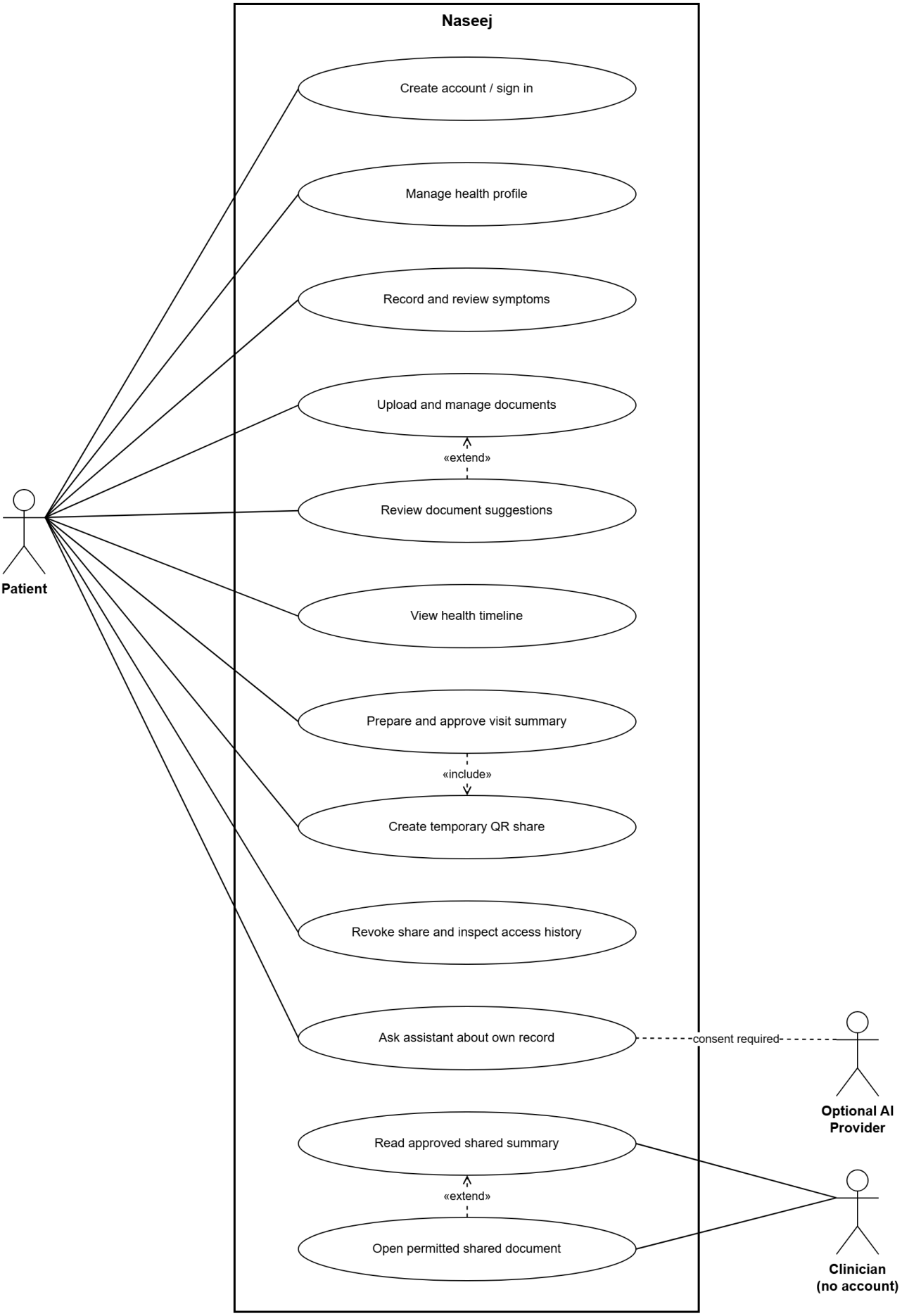
# 2. Flowchart: upload a document and review AI suggestions



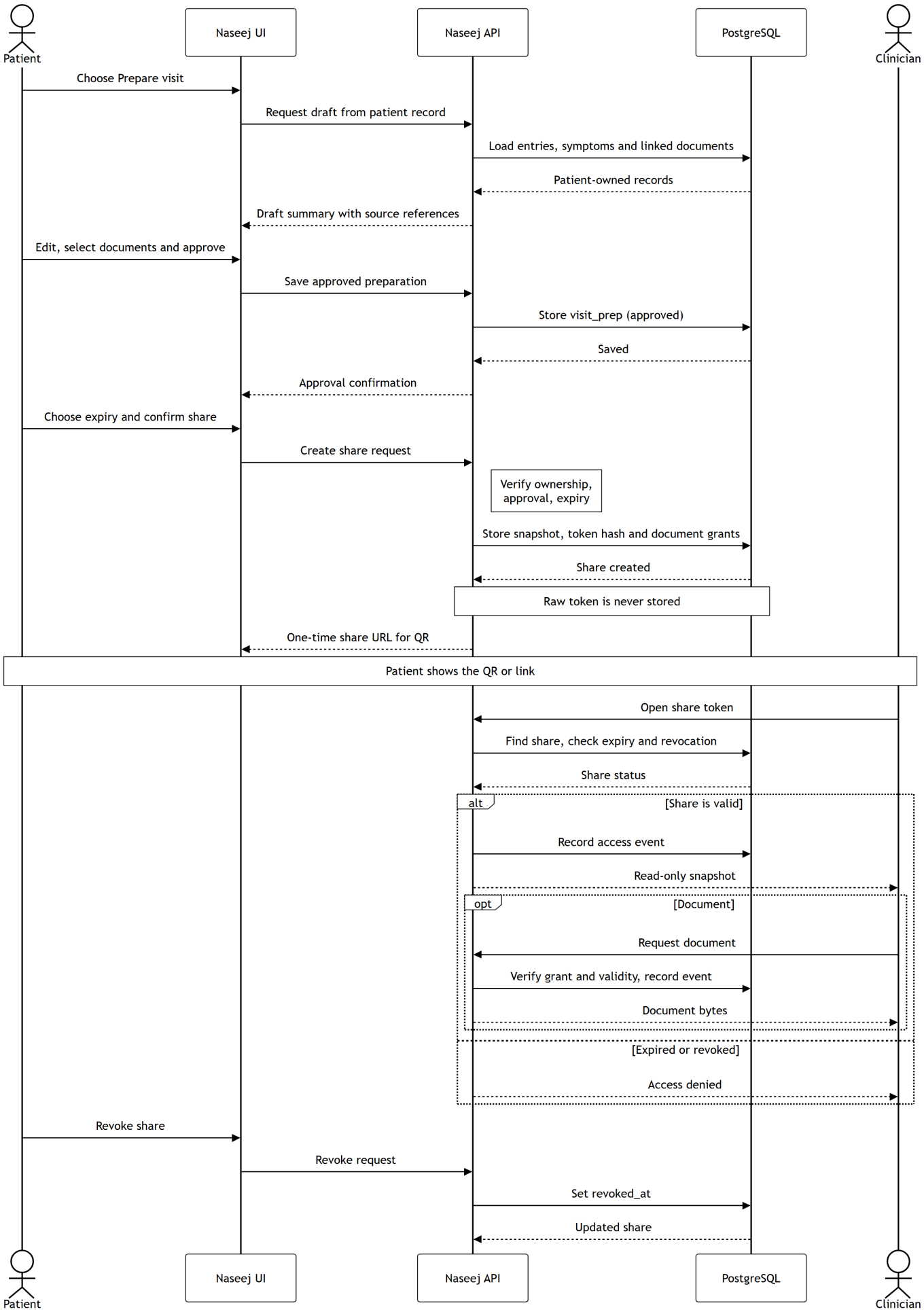
# 3. Activity diagram: prepare a visit summary and share it



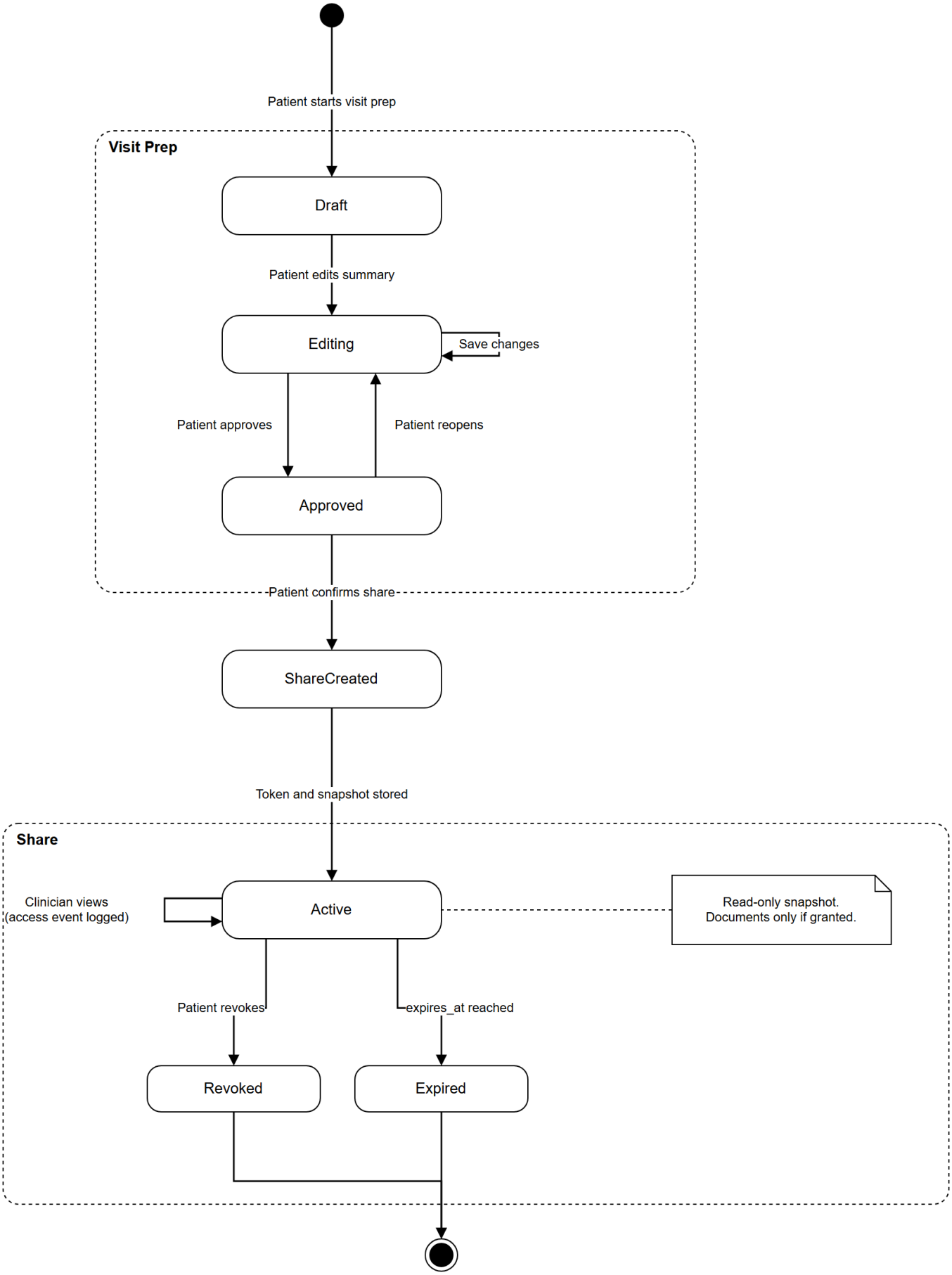
# 4. Use case diagram



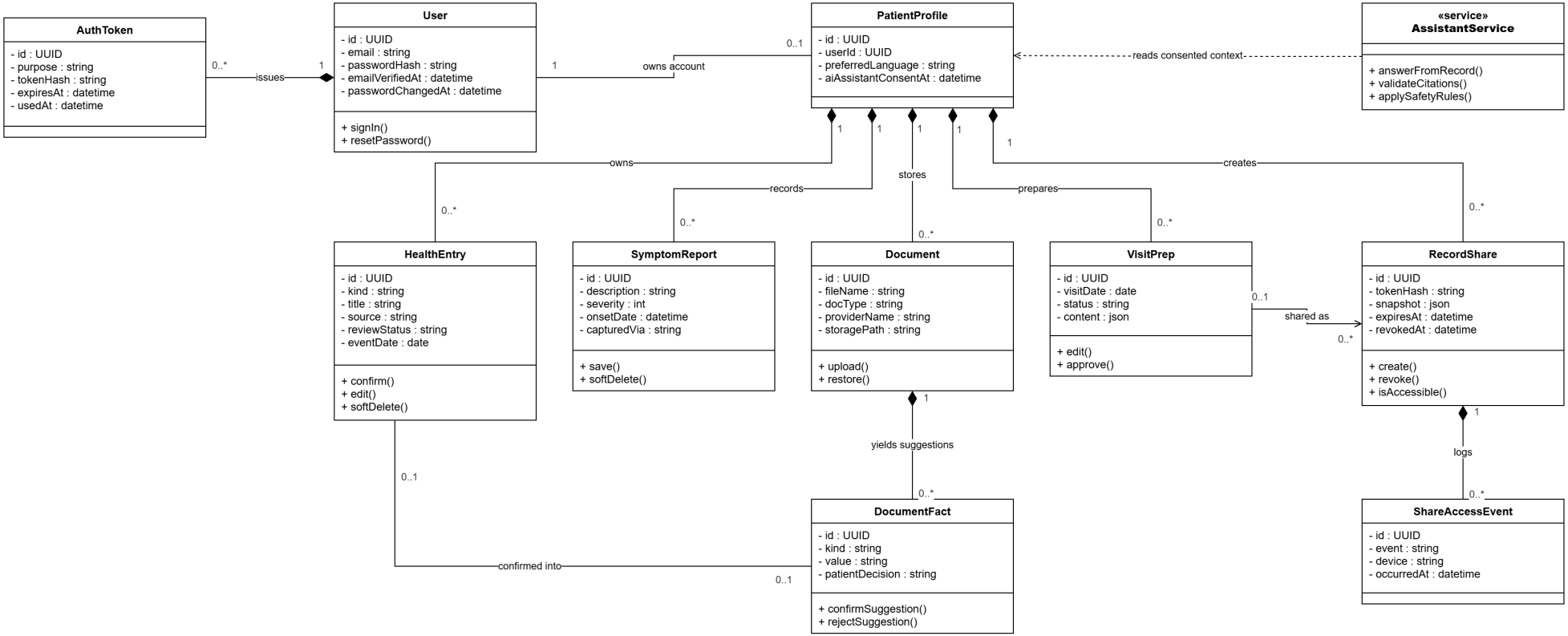
# 5. Sequence diagram: approve, share by QR, clinician reads, patient revokes



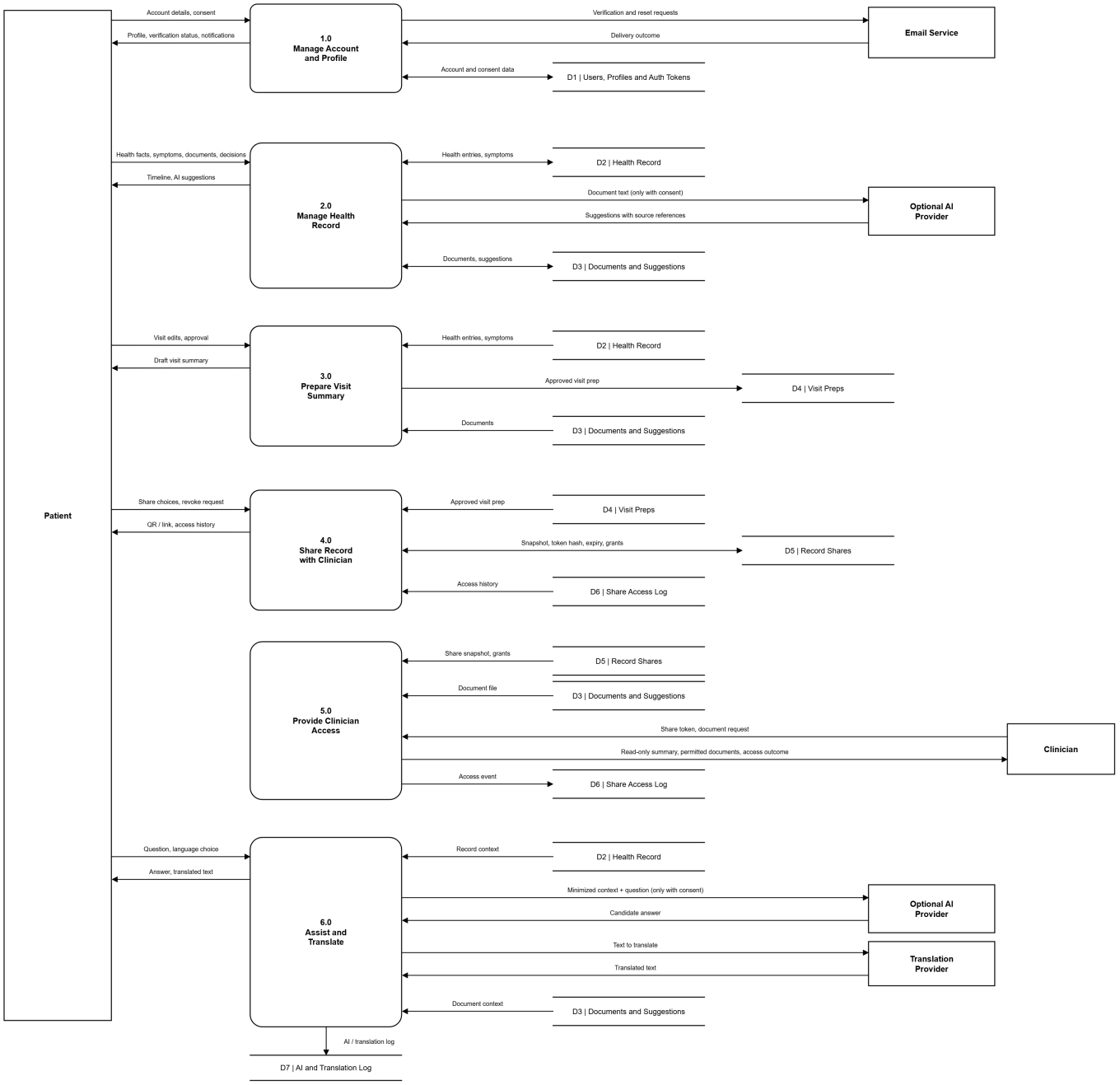
## 6. State diagram: visit preparation and share lifecycle



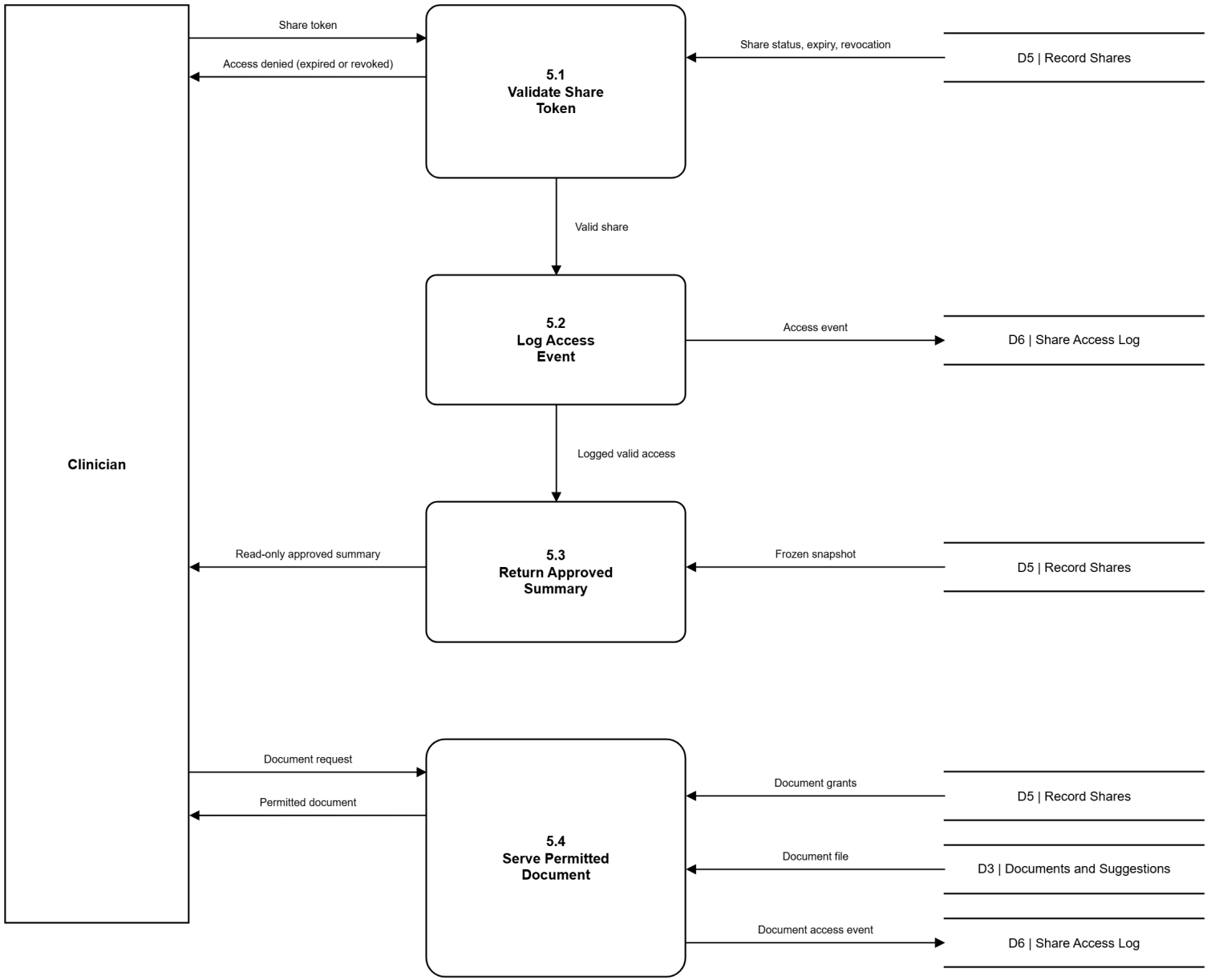
# 7. Class diagram



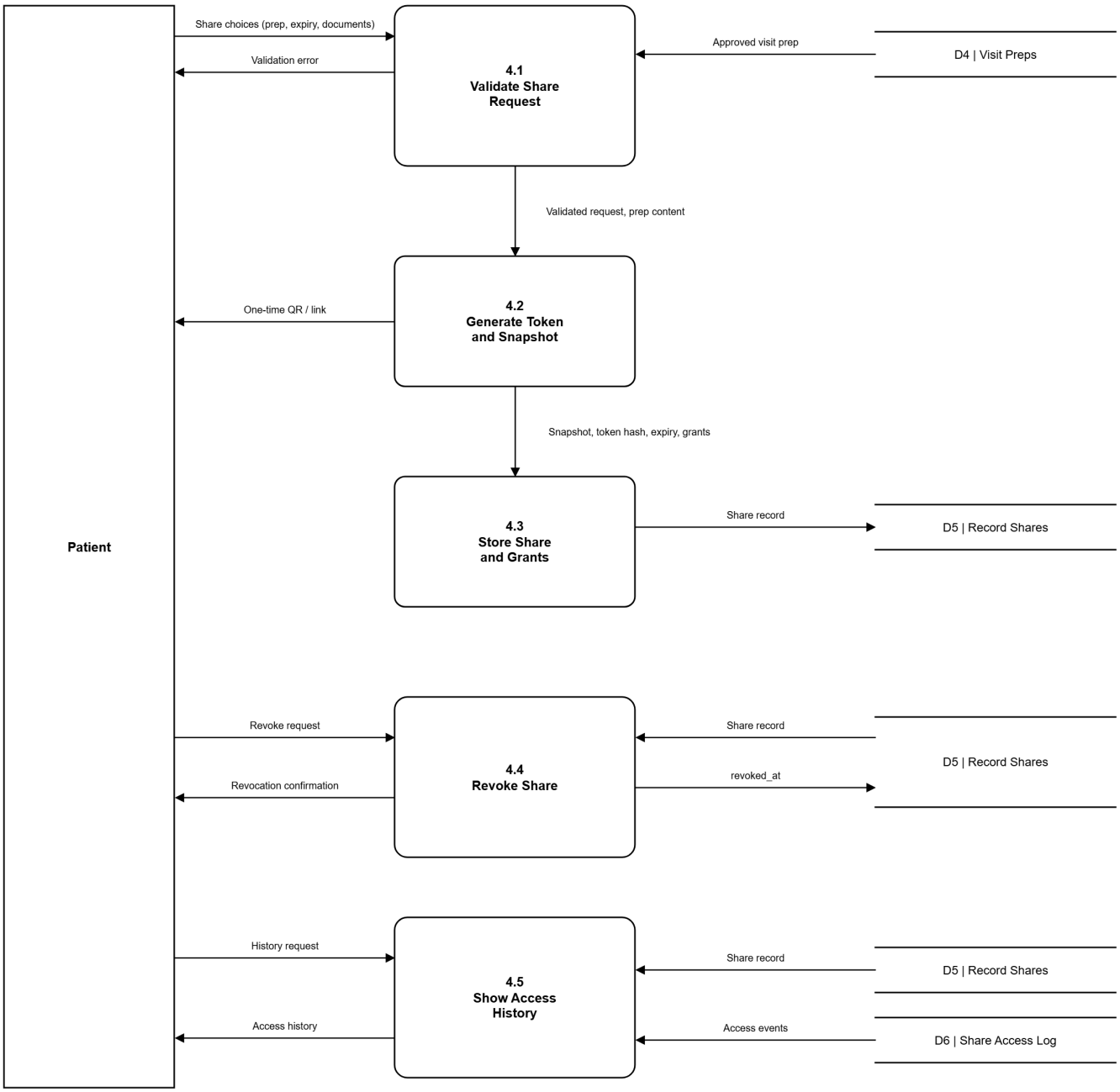
# 8. Data flow diagram, level 0



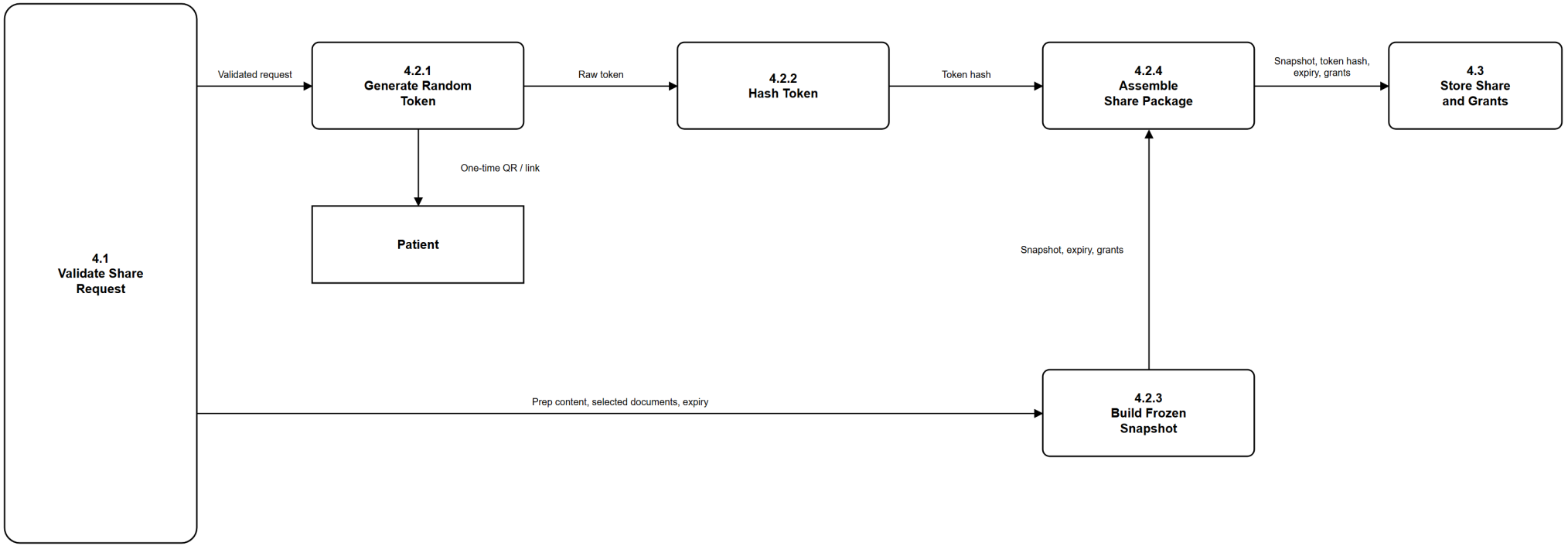
# 9. Data flow diagram, level 1: clinician access (5.0)



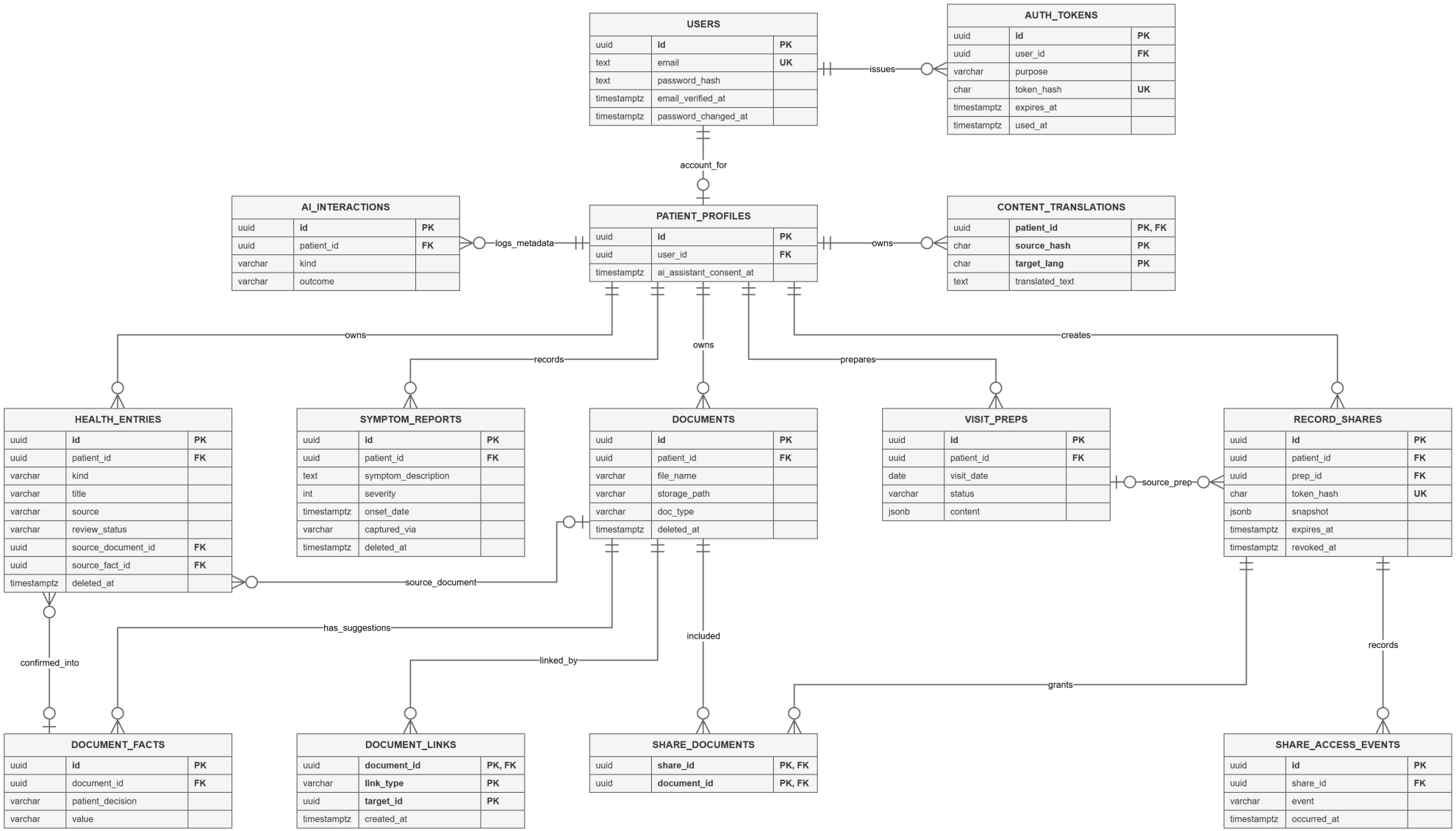
# 10. Data flow diagram, level 1: share record (4.0)



# 11. Data flow diagram, level 2: token and snapshot (4.2)



# 12. Entity relationship diagram (ERD)



# ERD to Relational Schema Mapping

Naseej patient health record platform | 14 tables | 17 relationships

## 1. Mapping rules applied

| ERD construct | Relational rule |
|---|---|
| Strong entity | Becomes one table; the entity identifier becomes the primary key (PK). |
| Attribute | Becomes one column; the data type is kept as drawn in the ERD. |
| Unique attribute | Column with a UNIQUE key (UK): USERS.email, AUTH_TOKENS.token_hash, RECORD_SHARES.token_hash. |
| One-to-many (1 : N) | The PK of the 1-side is copied into the N-side table as a foreign key (FK). |
| One-to-one, optional (1 : 0..1) | As 1 : N, with the FK also UNIQUE so at most one row points to the parent (PATIENT_PROFILES.user_id). |
| Optional participation (0..1) | The FK column accepts NULL (RECORD_SHARES.prep_id, HEALTH_ENTRIES.source_document_id, HEALTH_ENTRIES.source_fact_id). |
| Many-to-many (M : N) | Resolved by an associative table whose composite PK is made of the two FKs (SHARE_DOCUMENTS). |
| Dependent entity | The table takes the parent key as part of a composite PK (CONTENT_TRANSLATIONS, DOCUMENT_LINKS). |

## 2. Relational schema

Notation: **TABLE** ( primary key, *foreign key* → referenced table, *key that is both*, other columns ). [UK] marks a unique key. Only the columns shown in the ERD are listed.

**USERS** ( id, email [UK], password_hash, email_verified_at, password_changed_at )

**AUTH_TOKENS** ( id, *user_id* → USERS, purpose, token_hash [UK], expires_at, used_at )

**PATIENT_PROFILES** ( id, *user_id* → USERS [UK], ai_assistant_consent_at )

**AI_INTERACTIONS** ( id, *patient_id* → PATIENT_PROFILES, kind, outcome )

**CONTENT_TRANSLATIONS** ( *patient_id* → PATIENT_PROFILES, source_hash, target_lang, translated_text )

**HEALTH_ENTRIES** ( id, *patient_id* → PATIENT_PROFILES, kind, title, source, review_status, *source_document_id* → DOCUMENTS, *source_fact_id* → DOCUMENT_FACTS, deleted_at )

**SYMPTOM_REPORTS** ( id, *patient_id* → PATIENT_PROFILES, symptom_description, severity, onset_date, captured_via, deleted_at )

**DOCUMENTS** ( id, *patient_id* → PATIENT_PROFILES, file_name, storage_path, doc_type, deleted_at )

**DOCUMENT_FACTS** ( id, *document_id* → DOCUMENTS, patient_decision, value )

**DOCUMENT_LINKS** ( *document_id* → DOCUMENTS, link_type, target_id, created_at )

**VISIT_PREPS** ( id, *patient_id* → PATIENT_PROFILES, visit_date, status, content )

**RECORD_SHARES** ( id, *patient_id* → PATIENT_PROFILES, *prep_id* → VISIT_PREPS, token_hash [UK], snapshot, expires_at, revoked_at )

**SHARE_DOCUMENTS** ( *share_id* → RECORD_SHARES, *document_id* → DOCUMENTS )

**SHARE_ACCESS_EVENTS** ( id, *share_id* → RECORD_SHARES, event, occurred_at )

## 3. Relationship mapping

Each relationship drawn in the ERD and how it is realised as a foreign key. Cardinality is read parent : child.

| # | Parent | Child | Relationship | Cardinality | Mapping result |
|---|---|---|---|---|---|
| 1 | USERS | PATIENT_PROFILES | account_for | 1 : 0..1 | FK PATIENT_PROFILES.user_id → USERS.id, UNIQUE |
| 2 | USERS | AUTH_TOKENS | issues | 1 : 0..* | FK AUTH_TOKENS.user_id → USERS.id |
| 3 | PATIENT_PROFILES | AI_INTERACTIONS | logs_metadata | 1 : 0..* | FK AI_INTERACTIONS.patient_id → PATIENT_PROFILES.id |
| 4 | PATIENT_PROFILES | CONTENT_TRANSLATIONS | owns | 1 : 0..* | FK part of composite PK (patient_id, source_hash, target_lang) |
| 5 | PATIENT_PROFILES | HEALTH_ENTRIES | owns | 1 : 0..* | FK HEALTH_ENTRIES.patient_id → PATIENT_PROFILES.id |
| 6 | PATIENT_PROFILES | SYMPTOM_REPORTS | records | 1 : 0..* | FK SYMPTOM_REPORTS.patient_id → PATIENT_PROFILES.id |
| 7 | PATIENT_PROFILES | DOCUMENTS | owns | 1 : 0..* | FK DOCUMENTS.patient_id → PATIENT_PROFILES.id |
| 8 | PATIENT_PROFILES | VISIT_PREPS | prepares | 1 : 0..* | FK VISIT_PREPS.patient_id → PATIENT_PROFILES.id |
| 9 | PATIENT_PROFILES | RECORD_SHARES | creates | 1 : 0..* | FK RECORD_SHARES.patient_id → PATIENT_PROFILES.id |
| 10 | VISIT_PREPS | RECORD_SHARES | source_prep | 0..1 : 0..* | Optional FK RECORD_SHARES.prep_id → VISIT_PREPS.id (nullable) |
| 11 | DOCUMENTS | DOCUMENT_FACTS | has_suggestions | 1 : 0..* | FK DOCUMENT_FACTS.document_id → DOCUMENTS.id |
| 12 | DOCUMENTS | HEALTH_ENTRIES | source_document | 0..1 : 0..* | Optional FK HEALTH_ENTRIES.source_document_id → DOCUMENTS.id (nullable) |
| 13 | DOCUMENT_FACTS | HEALTH_ENTRIES | confirmed_into | 0..1 : 0..* | Optional FK HEALTH_ENTRIES.source_fact_id → DOCUMENT_FACTS.id (nullable) |
| 14 | DOCUMENTS | DOCUMENT_LINKS | linked_by | 1 : 0..* | FK part of composite PK (document_id, link_type, target_id) |
| 15 | DOCUMENTS | SHARE_DOCUMENTS | included | 1 : 0..* | M:N resolved; FK part of composite PK (share_id, document_id) |
| 16 | RECORD_SHARES | SHARE_DOCUMENTS | grants | 1 : 0..* | M:N resolved; FK part of composite PK (share_id, document_id) |
| 17 | RECORD_SHARES | SHARE_ACCESS_EVENTS | records | 1 : 0..* | FK SHARE_ACCESS_EVENTS.share_id → RECORD_SHARES.id |

## 4. Table specifications

PK = primary key, FK = foreign key (→ referenced table), UK = unique key. Data types are those shown in the ERD.

**USERS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| email | text | UK |
| password_hash | text | |
| email_verified_at | timestamptz | |
| password_changed_at | timestamptz | |

**AUTH_TOKENS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| user_id | uuid | FK → USERS |
| purpose | varchar | |
| token_hash | char | UK |
| expires_at | timestamptz | |
| used_at | timestamptz | |

**PATIENT_PROFILES**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| user_id | uuid | FK → USERS, UK |
| ai_assistant_consent_at | timestamptz | |

**AI_INTERACTIONS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| kind | varchar | |
| outcome | varchar | |

**CONTENT_TRANSLATIONS**

| Column | Type | Key |
|---|---|---|
| **patient_id** | uuid | PK, FK → PATIENT_PROFILES |
| **source_hash** | char | PK |
| **target_lang** | char | PK |
| translated_text | text | |

**HEALTH_ENTRIES**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| kind | varchar | |
| title | varchar | |
| source | varchar | |
| review_status | varchar | |
| source_document_id | uuid | FK → DOCUMENTS |
| source_fact_id | uuid | FK → DOCUMENT_FACTS |
| deleted_at | timestamptz | |

**SYMPTOM_REPORTS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| symptom_description | text | |
| severity | int | |
| onset_date | timestamptz | |
| captured_via | varchar | |
| deleted_at | timestamptz | |

**DOCUMENTS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| file_name | varchar | |
| storage_path | varchar | |
| doc_type | varchar | |
| deleted_at | timestamptz | |

**DOCUMENT_FACTS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| document_id | uuid | FK → DOCUMENTS |
| patient_decision | varchar | |
| value | varchar | |

**DOCUMENT_LINKS**

| Column | Type | Key |
|---|---|---|
| **document_id** | uuid | PK, FK → DOCUMENTS |
| **link_type** | varchar | PK |
| **target_id** | uuid | PK |
| created_at | timestamptz | |

**VISIT_PREPS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| visit_date | date | |
| status | varchar | |
| content | jsonb | |

**RECORD_SHARES**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| patient_id | uuid | FK → PATIENT_PROFILES |
| prep_id | uuid | FK → VISIT_PREPS |
| token_hash | char | UK |
| snapshot | jsonb | |
| expires_at | timestamptz | |
| revoked_at | timestamptz | |

**SHARE_DOCUMENTS**

| Column | Type | Key |
|---|---|---|
| **share_id** | uuid | PK, FK → RECORD_SHARES |
| **document_id** | uuid | PK, FK → DOCUMENTS |

**SHARE_ACCESS_EVENTS**

| Column | Type | Key |
|---|---|---|
| **id** | uuid | PK |
| share_id | uuid | FK → RECORD_SHARES |
| event | varchar | |
| occurred_at | timestamptz | |

**Scope note.** The ERD shows the active patient-owned model with its key columns only. The full schema (migrations 001 to 007, including the legacy multi-centre tables that remain unused) is described in docs/reference/database-erd.md.